

## Pairwise correlations among Vancouver vs suburban types

|           | VanSingle | VanCondo | FVSingle | FVCondo |
|-----------|-----------|----------|----------|---------|
| VanSingle | 1.00      | 0.71     | 0.82     | 0.54    |
| VanCondo  | 0.71      | 1.00     | 0.71     | 0.70    |
| FVSingle  | 0.82      | 0.71     | 1.00     | 0.71    |
| FVCondo   | 0.54      | 0.70     | 0.71     | 1.00    |

# Useful reference? Nihilism vs hedonism?

Berliant/McMillen

- ▶ Might make same linear gradient choice?

# So what would be helpful in intro/conclusion

Some ad hoc thoughts

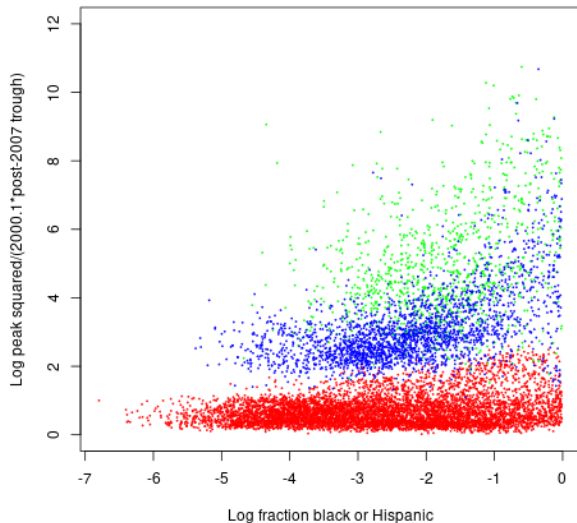
- ▶ What decision would be affected by different index choice?
- ▶ Portfolio efficiency for investor?
- ▶ Bailout decision for defaulters?
- ▶ Rationalize cap rate differences?

## Maybe overly bold claim

- ▶ “insufficient attention has been paid to the implications of non-random sampling within cities”
- ▶ Many people were thinking about defaults polluting indexes (or lack thereof biasing upward) during GFC
- ▶ If no pubs maybe because no one found anything?
- ▶ Vague recollection I tried and found nothing
- ▶ But location of course within metro.
- ▶ See my review of Handbook of Urban Econ thing ...

# Zip code price movements boom bust (source, Zillow)

Red: Flyover, Blue: Coastal, green: Sand State



# Price of services

“If the objective is to measure the price of housing services or housing space, then the definitions of hand  $H$  in the preceding equations should reflect a measure of housing services, rather than assuming that each housing unit provides the same quantity of housing space.”

- Is the point that Bill Gates's house should get extra weight?

# Muth discussion

- ▶ What is going on with  $\tau k$  and  $\frac{\partial \tau}{\partial k}$  Linear? Nonlinear?
- ▶ Per above, linear in distance not realistic due to congestion.  
Going 1 mile on Cross-Bronx Expressway slower than one mile on 95 in the suburbs.

# Common appreciation

- ▶ Standard monocentric gradient linear
  - ▶ Value at  $r$  = structure + lot size  $\theta \int_{s=r}^b \frac{1}{\text{lot size at } s} ds$
  - ▶ Time log derivative unlikely constant by  $r$
- ▶ But no less realistically (per Williams)  
 $\log v = \log \text{structure} + \theta [\ln b(t) - \ln r]$ 
  - ▶ Then location not clear a source of growth heterogeneity
  - ▶ Might be fine with, say unit income elasticity
- ▶ Zoned single family vs multifamily clearer?



## Proposition 2 doesn't really follow from Prop 1

- ▶ Are you missing “if not quasi-linearity, and if linear gradient in distance”?

# Is AHS price data bananas?

- ▶ 2015-2019 Table 1 shows 18% vs 13% per year appreciation
- ▶ FHFA per FRED says about 7%
- ▶ Washington, D.C. Table 1 says about 2%
- ▶ FHFA per FRED says about 8%.
- ▶ Even I can afford Attom . . .
- ▶ Aggregate Zillow zip indexes by metro central/suburbs?
- ▶ Given giant idiosyncratic errors, what can we rule out?
- ▶ Strong suggestion: use better data and see if differences as large